CRTBP Collision manifold integration notes

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1 Choosing the best coordinates

We fix our parameter, μ, to be the mass of the small primary with is located at the position $(\mu-1,0)$. So we have $0<\mu\leq\frac{1}{2}$ and similarly, the large primary has mass $1-\mu$ and its position in configuration space is $(\mu,0)$.

1.1 The 3 vector fields

Once and for all we write down the formulas for all 3 vector fields. Here $\mu_2=\mu$ and $\mu_1=1-\mu_2$ in any case they appear. We let f_0,f_1,f_2 denote the CRTBP vector field, the regularized field at the large primary, and the regularized field at the small primary respectively.

1.1.1 Standard

The Szebehely version of the CRTBP has mass parameter $0<\mu\leq\frac{1}{2}$. The large primary is located at $P_1=(\mu,0)$ having mass, $1-\mu$ and the small primary is located at $P_2=(\mu-1,0)$ having mass, μ. The vector field prior to automatic differentiation is given by the equations

$$\begin{aligned}
\dot{x}&=p\\
\dot{p}&=2q+(1-\mu)(x-\mu)+\mu(x+1-\mu)-\frac{(1-\mu)(x-\mu)}{d_1^3}-\frac{\mu(x+1-\mu)}{d_2^3}\\
\dot{y}&=q\\
\dot{q}&=-2p+y-\frac{(1-\mu)y}{d_1^3}-\frac{\mu y}{d_2^3}
\end{aligned}$$

where d_1,d_2 are defined by the formulas

$$d_1=\sqrt{(x-\mu)^2+y^2}\qquad d_2=\sqrt{(x+1-\mu)^2+y^2}$$

Below, we will often write $\mu_1=1-\mu$ and $\mu_2=\mu$.

The (extended) standard CRTBP has coordinates are denoted by $(x_0,p_0,y_0,q_0,r_0,s_0))$. The vector field is denoted by, $f_0:\mathbb{R}^6\to\mathbb{R}^6$, is obtained after automatic differentiation by setting $r_0=d_1^{-1}$ and $s_0=d_2^{-1}$. The explicit formulas are now given by the polynomial

formulas

$$
\begin{aligned}
\dot{x}_0 &= p_0 \\
\dot{p}_0 &= 2q_0 + \mu_1 (x_0 - \mu_2) + \mu_2 (x_0 + \mu_1) - \mu_1 (x_0 - \mu_2) r_0^3 - \mu_2 (x_0 + \mu_1) s_0^3 \\
\dot{y}_0 &= q_0 \\
\dot{q}_0 &= -2p_0 + \mu_1 y_0 + \mu_2 y_0 - \mu_1 y_0 r_0^3 - \mu_2 y_0 s_0^3 \\
\dot{r_0} &= -r_0^3 \left((x_0 - \mu_2) p_0 + y_0 q_0 \right) \\
\dot{s_0} &= -s_0^3 \left((x_0 + \mu_1) p_0 + y_0 q_0 \right)
\end{aligned}
$$

where r_0 and s_0 are the automatic differentiation coordinates given by

$$
r_0 = \frac{1}{\sqrt{(x_0 - \mu_2)^2 + y_0^2}} \qquad s_0 = \frac{1}{\sqrt{(x_0 + \mu_1)^2 + y_0^2}}.
$$

After factoring, it is often convenient to write the formulas for $\dot{p}_0, \dot{q}_0$ as

$$
\begin{aligned}
\dot{p}_0 &= 2q_0 + \mu_1 (x - \mu_2) \left(1 - r_1^3\right) + \mu_2 (x + \mu_1) \left(1 - r_2^3\right) \\
\dot{q}_0 &= -2p_0 + \mu_1 y \left(1 - r_1^3\right) + \mu_2 y \left(1 - r_2^3\right)
\end{aligned}
$$

1.1.2 Large primary regularization

The vector field $f_1 : \mathbb{R}^5 \to \mathbb{R}^5$ is the vector field desingularized at the large primary located at coordinates $(\mu, 0)$ in configuration space. It provides the derivatives given by the formulas

$$
\begin{aligned}
\dot{x}_1 &= p_1 \\
\dot{p}_1 &= 8 \left(x_1^2 + y_1^2\right) q_1 + 12 x_1 \left(x_1^2 + y_1^2\right)^2 + 16\mu x_1^3 + 4(\mu - C) x_1 + 8\mu x_1 \left(x_1^2 - 3y_1^2 + 1\right) r_1^3 \\
\dot{y}_1 &= q_1 \\
\dot{q}_1 &= -8 \left(x_1^2 + y_1^2\right) p_1 + 12 y_1 \left(x_1^2 + y_1^2\right)^2 - 16\mu y_1^3 + 4(\mu - C) y_1 + 8\mu y_1 \left(-y_1^2 + 3x_1^2 + 1\right) r_1^3 \\
\dot{r}_1 &= -2r_1^3 \left(\left(x_1^2 + y_1^2\right) (x_1 p_1 + y_1 q_1) + (x_1 p_1 - y_1 q_1) \right)
\end{aligned}
$$

where $C = H_0(x_0, p_0, y_0, q_0)$ is the standard energy and r_1 is given by

$$
r_1 = \frac{1}{\sqrt{(x_1^2 + y_1^2)^2 + 1 + 2(x_1^2 - y_1^2)}}.
$$

1.1.3 Small primary regularization

The vector field $f_2 : \mathbb{R}^5 \to \mathbb{R}^5$ is the vector field desingularized at the small primary located at coordinates $(\mu - 1, 0)$. It provides the derivatives given by the formulas

$$
\begin{aligned}
\dot{x}_2 &= p_2 \\
\dot{p}_2 &= 8 \left(x_2^2 + y_2^2\right) q_2 + 12 x_2 \left(x_2^2 + y_2^2\right)^2 + 16(\mu - 1) x_2^3 + 4(1 - \mu - C) x_2 + 8(1 - \mu) x_2 \left(-x_2^2 + 3y_2^2 + 1\right) r_2^3 \\
\dot{y}_2 &= q_2 \\
\dot{q}_2 &= -8 \left(x_2^2 + y_2^2\right) p_2 + 12 y_2 \left(x_2^2 + y_2^2\right)^2 - 16(\mu - 1) y_2^3 + 4(1 - \mu - C) y_2 + 8(1 - \mu) y_2 \left(y_2^2 - 3x_2^2 + 1\right) r_2^3 \\
\dot{r}_2 &= -2r_2^3 \left(\left(x_2^2 + y_2^2\right) (x_2 p_2 + y_2 q_2) - (x_2 p_2 - y_2 q_2) \right)
\end{aligned}
$$

where $C = H_0(x_0, p_0, y_0, q_0)$ is the standard energy and r_2 is given by

$$
r_2 = \frac{1}{\sqrt{(x_2^2 + y_2^2)^2 + 1 + 2(y_2^2 - x_2^2)}}.
$$

1.2 Changing coordinates

1.3 Mappings between coordinates

The basis for these coordinate transformations are the mappings $z \mapsto w^2 + \mu$ and $z \mapsto w^2+\mu-1$ which define the desingularized coordinates about the large and small primaries respectively.

1.3.1 standard to large primary

For the regularized vector field, f_1 (desingularized at the large primary) we have the coordinates (x_1, p_1, y_1, q_1) which correspond to the standard CRTBP coordinates (x_0, p_0, y_0, q_0) by the formulas:

$$\begin{aligned} x_0 &= x_1^2 - y_1^2 + \mu \\ p_0 &= \frac{x_1 p_1 - y_1 q_1}{2(x_1^2 + y_1^2)} \\ y_0 &= 2x_1 y_1 \\ q_0 &= \frac{y_1 p_1 + x_1 q_1}{2(x_1^2 + y_1^2)} \end{aligned}$$

Note that p_0, q_0 are the t-time derivatives of x_0, y_0 while p_1, y_1 are the τ-time derivatives of x_1, y_1 and the time rescaling is related by

$$\frac{dt}{d\tau} = 4w\overline{w} \qquad \frac{d\tau}{dt} = \frac{1}{4w\overline{w}}$$

where $w\overline{w} = x_1^2 + y_1^2$. Then we have for example from $z = x_0 + iy_0 = w^2 + \mu$, the formula

$$\begin{aligned} \frac{dz}{dt} &= 2w\frac{dw}{d\tau}\frac{d\tau}{dt} \\ &= 2(x_1 + iy_1)(p_1 + iq_1)(\frac{1}{4w\overline{w}}) \\ &= \frac{x_1 p_1 - y_1 q_1}{2(x_1^2 + y_1^2)} + i\frac{x_1 q_1 + y_1 p_1}{2(x_1^2 + y_1^2)} \end{aligned}$$

and the values of p_0, q_0 follow from noting that $\dot{z} = \dot{x}_0 + i\dot{y}_0$.

These formulas can be solved in the other direction to obtain the regularization transformation

$$\begin{aligned} x_1 &= \mathrm{Re}\left(\sqrt{x_0 + iy_0 - \mu}\right) \\ p_1 &= 2(p_0 x_1 + q_0 y_1) \\ y_1 &= \mathrm{Im}\left(\sqrt{x_0 + iy_0 - \mu}\right) \\ q_1 &= 2(q_0 x_1 - p_0 y_1) \end{aligned}$$

Again, this computation follows from the computation

$$\frac{d}{d\tau}(w^2) = 2w\frac{dw}{d\tau} = \frac{d}{d\tau}(z - \mu) = \frac{dz}{dt}\frac{dt}{d\tau}.$$

So we obtain

$$\begin{aligned}\frac{dw}{d\tau} &= \frac{4\dot{z}w\overline{w}}{2w} \\ &= 2\dot{z}\overline{w} \\ &= 2(p_0x_1 + q_0y_1) + 2i(q_0x_1 - p_0y_1).\end{aligned}$$

The potential in the new coordinates is

$$\mathcal{U}_1(x_1, y_1) = 2(x_1^2+y_1^2)^3+4\mu(x_1^4-y_1^4)+2(\mu-C)(x_1^2+y_1^2)+4(1-\mu)+\frac{4\mu(x_1^2+y_1^2)}{\sqrt{(x_1^2+y_1^2)^2+1+2(x_1^2-y_1^2)}}.$$

which gives rise to the the Jacobi constant in the new coordinates which is

$$H_1(x_1, p_1, y_1, q_1) = 2\mathcal{U}_1(x_1, y_1) - p_1^2 - q_1^2.$$

1.3.2 standard to small primary

Working similar to the previous computations, we have formulas for the small body regularization.

$$\begin{aligned}x_0 &= x_2^2 - y_2^2 + \mu - 1 \\ p_0 &= \frac{x_2p_2 - y_2q_2}{2(x_2^2+y_2^2)} \\ y_0 &= 2x_2y_2 \\ q_0 &= \frac{x_2q_2 + y_2p_2}{2(x_2^2+y_2^2)}\end{aligned}$$

These formulas can be solved in the other direction to obtain the regularization transformation

$$\begin{aligned}x_2 &= \mathrm{Re}\left(\sqrt{x_0 + iy_0 - (\mu-1)}\right) \\ p_2 &= 2\left(p_0x_2 + q_0y_2\right) \\ y_2 &= \mathrm{Im}\left(\sqrt{x_0 + iy_0 - (\mu-1)}\right) \\ q_2 &= 2\left(q_0x_2 - p_0y_2\right)\end{aligned}$$

The potential in the new coordinates is

$$\mathcal{U}_2(x_2, y_2) = 2(x_2^2+y_2^2)^3+4(\mu-1)(x_2^4-y_2^4)+2(1-\mu-C)(x_2^2+y_2^2)+4\mu+\frac{4(1-\mu)(x_2^2+y_2^2)}{\sqrt{(x_2^2+y_2^2)^2+1+2(x_2^2-y_2^2)}}.$$

which gives rise to the the Jacobi constant in the new coordinates which is

$$H_2(x_2, p_2, y_2, q_2) = 2\mathcal{U}_2(x_2, y_2) - p_2^2 - q_2^2.$$

We summarize and denote these changes of coordinates in a vectorized form for further reference. Begin by defining three Euclidean spaces, $X_i \cong \mathbb{R}^4$ for $i = 0, 1, 2$. Of course these spaces are isomorphic, however, we will write $v \in X_i$ when referring to points which are being integrated by the corresponding vector field, f_i. In other words, X_i will be used

to denote the phase space for the dynamics generated by f_i. We define maps between these spaces, $\mathcal{F}_i : X_0 \to X_i$ for $i = 1, 2$ given by

$$(x_0, p_0, y_0, q_0) \overset{\mathcal{F}_i}{\longmapsto} (x_i, p_i, y_i, q_i)$$

and similarly, the inverse maps, $F_i^{-1} : X_i \to X_0$ are defined

$$(x_i, p_i, y_i, q_i) \overset{\mathcal{F}_i^{-1}}{\longmapsto} (x_0, p_0, y_0, q_0).$$

1.3.3 small to large primary and back

These maps naturally compose to produce a map between regularized coordinates. For example, changing from f_2 to f_1 coordinates corresponds to the map

$$(x_1, p_1, y_1, q_1) = \mathcal{F}_1 \circ \mathcal{F}_2^{-1} (x_2, p_2, y_2, q_2) .$$

Using the above formulas for $\mathcal{F}_1, \mathcal{F}_2$, we can write the formulas for $\mathcal{F}_1 \circ \mathcal{F}_2^{-1}$ which is given by

$$\begin{aligned}
x_1 &= \mathrm{Re}\left(\sqrt{x_2^2 - y_2^2 - 1 + i(2x_2y_2)}\right) \\
p_1 &= \frac{x_1(x_2p_2 - y_2q_2) + y_1(x_2q_2 + y_2p_2)}{x_2^2 + y_2^2} \\
y_1 &= \mathrm{Im}\left(\sqrt{x_2^2 - y_2^2 - 1 + i(2x_2y_2)}\right) \\
q_1 &= \frac{x_1(x_2q_2 + y_2p_2) - y_1(x_2p_2 - y_2q_2)}{x_2^2 + y_2^2}
\end{aligned}$$

Likewise, we obtain formulas for the map from f_1 coordinates to f_2 coordinates given by $\mathcal{F}_2 \circ \mathcal{F}_1^{-1}$ given by

$$\begin{aligned}
x_2 &= \mathrm{Re}\left(\sqrt{x_1^2 - y_1^2 + 1 + i(2x_1y_1)}\right) \\
p_2 &= \frac{x_2(x_1p_1 - y_1q_1) + y_2(x_1q_1 + y_1p_1)}{x_1^2 + y_1^2} \\
y_2 &= \mathrm{Im}\left(\sqrt{x_1^2 - y_1^2 + 1 + i(2x_1y_1)}\right) \\
q_2 &= \frac{x_2(x_1q_1 + y_1p_1) - y_2(x_1p_1 - y_1q_1)}{x_1^2 + y_1^2}
\end{aligned}$$

1.3.4 Normalizing time

In order to keep a global track of time for orbits with respect to their position in X_0, we will normalize the time along orbits in regularized coordinates. We let $t \in \mathbb{R}$ always denote this time and use τ_1, τ_2 respectively, to keep time along orbits when integrating in X_1, X_2 respectively. This normalization must satisfy the following property. Suppose $v \in X_0$ is an arbitrary initial condition. Let γ_0 denote the trajectory through v satisfying $\gamma_0(0) = v$. Now, let γ_i denote the trajectory segment for f_i satisfying $\gamma_i(0) = \mathcal{F}_i(v)$. Then, the normalized time along γ_i is the mapping, $t : \mathbb{R} \to \mathbb{R}$ satisfying

$$\gamma_0(t(\tau)) = \mathcal{F}_i^{-1} (\gamma_i(\tau)) \qquad \text{for all } \tau \in \mathbb{R}.$$

Recalling the coordinate changes defined $\mathcal{F}_i$ above, we have $\frac{dt}{d\tau_i} = 4w_i\overline{w}_i$. Noting that $w_i\overline{w}_i = |w_i|^2 = x_i^2 + y_i^2$, we can directly solve this differential equation to obtain the formula

$$t(\tau) = 4\int_0^\tau x_i(\tau_i)^2 + y_i(\tau_i)^2 \; d\tau_i.$$

Similarly, we have $\frac{d\tau_i}{dt} = \frac{1}{4w_i\overline{w}_i}$

1.4 When to change coordinates

We are interested in propagating material surfaces by dynamically switching to the coordinates which have the best properties. In our case, we would like to obtain the largest surface area per time-step for the Taylor integrator and this amounts to remaining bounded away from the singular points of each vector field. An analogous property which is easy to measure is the (squared) *speed* which we define pointwise in any coordinates as $S_i = p_i^2 + q_i^2$ for $i = 0, 1, 2$.

A reasonable strategy then is to integrate using whichever of the 3 vector fields has the minimum value for this speed along "most" of the boundary chart. To determine this, we compare $S_1 = p_1^2 + q_1^2$ in the (large primary) regularized coordinates, to the corresponding speed in the standard CRTBP coordinates, which is given explicitly by

$$\begin{aligned} S_0 = p_0^2 + q_0^2 &= \frac{(x_1p_1 - y_1q_1)^2 + (y_1p_1 + x_1q_1)^2}{4(x_1^2 + y_1^2)^2} \\ &= \frac{(x_1p_1)^2 + (y_1q_1)^2 + (y_1p_1)^2 + (x_1q_1)^2}{4(x_1^2 + y_1^2)^2} \\ &= \frac{(x_1^2 + y_1^2)(p_1^2 + q_1^2)}{4(x_1^2 + y_1^2)^2} \\ &= \frac{(p_1^2 + q_1^2)}{4(x_1^2 + y_1^2)} \end{aligned}$$

Evidently, we have $S_0 = S_1$ if and only if $x_1^2 + y_1^2 = \frac{1}{4}$ which is a circle of radius $\frac{1}{2}$ centered at the origin in the (x_01y_1) coordinates. We call this circle the *ideal* domain for f_1. Hence, for any points inside this circle we have $S_1 < S_0$ and we integrate using the regularized vector field. Otherwise, we switch to the standard coordinates before integrating and the formula $x_1^2 + y_1^2 = \frac{1}{4}$ provides the criteria to be checked when determining whether or not to change coordinates.

Similarly, if we are in f_0 coordinates, we would like a formula for determining when to change to f_1 coordinates. For this we compute the image of the ideal circle for f_1 using the formulas for x_0, y_0 above. In anticipation that this domain is a circle centered at $(\mu, 0)$ in the (x_0, y_0) plane, we compute

$$\begin{aligned} (x_0 - \mu)^2 + y_0^2 &= (x_1^2 - y_1^2)^2 + (2x_1y_1)^2 \\ &= (x_1^2 + y_1^2)^2 \\ &= \frac{1}{16} \end{aligned}$$

where in the last step we used the assumption that $x_1^2 + y_1^2 = \frac{1}{4}$. Thus, the ideal domain for which $S_0 < S_1$ is the interior of the disc bounded by the circle of radius $\frac{1}{4}$ centered at the large primary in the (x_0, y_0) plane. We use the formula, $(x_0 - \mu)^2 + y_0^2 = \frac{1}{16}$ jas the computable criteria for switching from f_0 coordinates into f_1 coordinates.

A similar computation shows that for points written in f_2 coordinates (near the small primary) there is a disc of radius $\frac{1}{2}$ given by the formula $x_2^2 + y_2^2 = \frac{1}{4}$ on which $S_2 < S_0$. This is the ideal domain for f_2 and we flow using this vector field inside of this disc, and otherwise, we switch to the f_0 coordinates. The corresponding condition for switching from f_0 into f_2 coordinates is given by the boundary circle $(x_0 - \mu + 1)^2 + y_0^2 = \frac{1}{16}$.

1.5 Integration details

Consider an arbitrary arc of initial data $\gamma : [-1, 1] \to \mathbb{C}^5$ with respect to, f_i, one of the regularized vector fields. We assume it is analytic with Taylor expansion given by

$$\gamma(\sigma) = \sum_{n=0}^{\infty} a_n \sigma^n \qquad a_n \in \mathbb{C}^5, \quad \sigma \in [-1, 1].$$

We also denote the equivalent coordinate-wise representation as follows.

$$\gamma(\sigma) = \begin{pmatrix} x_i(\sigma) \\ p_i(\sigma) \\ y_i(\sigma) \\ q_i(\sigma) \\ r_i(\sigma) \end{pmatrix} = \sum_{n=0}^{\infty} \begin{pmatrix} a_n^{(1)} \\ a_n^{(2)} \\ a_n^{(3)} \\ a_n^{(4)} \\ a_n^{(5)} \end{pmatrix} \sigma^n$$

That is, x_i, p_i, y_i, q_i, r_i are considered as one variable power series. In this work, these parameterize portions of collision manifolds for one of the primary bodies, or invariant manifolds attached to one of the equilibria. For computational purposes, these are represented in the computer as truncated coefficient sequence which defines a polynomial, and rigorous error bounds for the analytic tail i.e. an ℓ_1 error bound on the truncation. We denote this by the representation $\gamma(\sigma) = (\bar{\gamma}(\sigma), r)$ where $\bar{\gamma} : \mathbb{R} \to \mathbb{C}^5$ is a polynomial of order $N \geq 1$ and $r > 0$ satisfies the bound

$$\max \left\{ \sum_{N+1}^{\infty} \left| a_n^{(1)} \right|, \sum_{N+1}^{\infty} \left| a_n^{(2)} \right|, \sum_{N+1}^{\infty} \left| a_n^{(3)} \right|, \sum_{N+1}^{\infty} \left| a_n^{(4)} \right|, \sum_{N+1}^{\infty} \left| a_n^{(5)} \right| \right\} < r < \infty.$$

1.6 configuration space

We start with the configuration variables, x, y. We start by assuming $u(\sigma) = \bar{u}(\sigma) + h_u(\sigma)$ and $v(\sigma) = \bar{v}(\sigma) + h_v(\sigma)$ where $\bar{u}, \bar{v}$ are the polynomial parts and $h_u, h_v \in C^\omega(\mathbb{D})$ satisfy

$$\|\mathcal{T}(h_u)\|_{\ell_1} < r \qquad \|\mathcal{T}(h_v)\|_{\ell_1} < r.$$

From the formula for x we have $x(\sigma) = u(\sigma)^2 - v(\sigma)^2 + \mu_2$ so the polynomial part is obtained as the vector $\bar{x} = (x_0, \ldots, x_N)$ as the first $N + 1$ terms of we have the bound for the Taylor transform of x given by

$$\mathcal{T}(x) = ((u_0, \ldots, u_N) + \mathcal{T}(h_u))^2 + ((u_0, \ldots, u_N) + \mathcal{T}(h_u))^2$$

2 Regularized Jacobi Constant

We also compute for later reference, the regularized energy function. Given the original energy and potential functions

$$E(x, p, y, q) = \frac{1}{2}(p^2 + q^2) - \Omega(x, y) \qquad \Omega(x, y) = \frac{1}{2}(x^2 + y^2) + \frac{\mu_1}{r_0} + \frac{\mu_2}{s_0}.$$

In the regularized (at μ_1) coordinates we have $\mu = \mu_2$ so the potential is

$$\mathcal{U}(u,v) = 2\left(u^2+v^2\right)^3 + 4\mu\left(u^4-v^4\right) + 2(\mu - C)\left(u^2+v^2\right) + 4(1-\mu) + \frac{4\mu\left(u^2+v^2\right)}{\sqrt{\left(u^2+v^2\right)^2+1+2\left(u^2-v^2\right)}}.$$

We set $k_1 = u^2+v^2$ and $k_2 = u^2-v^2$ and rewrite this potential as

$$\mathcal{U}(u,v) = 2k_1^3 + 4\mu k_1 k_2 + 2(\mu - C)k_1 + 4(1-\mu) + \frac{4\mu k_1}{\sqrt{k_1^2+1+2k_2}}.$$

Now, we have the formulas for the partial derivatives

$$\begin{aligned}\frac{\partial \mathcal{U}}{\partial u} &= 12\left(u^2+v^2\right)^2 u + 16\mu u^3 + 4(\mu - C)u + \frac{8\mu u(u^2-3v^2+1)}{\left(\left(u^2+v^2\right)^2+2(u^2-v^2)+1\right)^{\frac{3}{2}}} \\ &= 12k_1^2 u + 16\mu u^3 + 4(\mu - C)u + \frac{8\mu u(u^2-3v^2+1)}{(k_1^2+2k_2+1)^{\frac{3}{2}}}\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \mathcal{U}}{\partial u} &= 12\left(u^2+v^2\right)^2 v - 16\mu v^3 + 4(\mu - C)v + \frac{8\mu v(3u^2-v^2+1)}{((u^2+v^2)^2+2(u^2-v^2)+1)^{\frac{3}{2}}} \\ &= 12k_1^2 v - 16\mu v^3 + 4(\mu - C)v + \frac{8\mu v(3u^2-v^2+1)}{(k_1^2+2k_2+1)^{\frac{3}{2}}}\end{aligned}$$

The Jacobi constant is now given by the formula

$$E(u,p,v,q) = p^2 + q^2 - 2\mathcal{U}(u,v)$$

We can evaluate $\frac{dE}{dt}$ along a trajectory using the formulas from the regularize vector field

$$\begin{aligned}\dot{u} &= p \\ \dot{p} &= 8k_1 q + \frac{\partial \mathcal{U}}{\partial u} \\ \dot{v} &= q \\ \dot{q} &= -8k_1 p + \frac{\partial \mathcal{U}}{\partial v}.\end{aligned}$$

Now, we introduce the variables $D = \sqrt{k_1^2+2k_2+1}$, $h_1 = pu+qv$ and $h_2 = pu-qv$ and simplifying with k_i, h_i, D we have

2.1 Formulas for kinetic energy

Lets compute $p\dot{p}$ and $q\dot{q}$.

$$\begin{aligned}p\dot{p} &= 8k_1 qp + 12k_1^2 up + 16\mu u^3 p + 4(\mu - C)up + \frac{8\mu up}{D^3}\left(u^2-3v^2+1\right) \\ q\dot{q} &= -8k_1 pq + 12k_1^2 vq - 16\mu v^3 q + 4(\mu - C)vq + \frac{8\mu vq}{D^3}\left(3u^2-v^2+1\right))\end{aligned}$$

and adding them together we get

$$p\dot{p} + q\dot{q} = 12k_1^2 h_1 + 16\mu(u^3 p - v^3 q) + 4(\mu - C)h_1 + \frac{8\mu}{D^3}\left(u^3 p - 3uv^2 p + 3u^2 vq - v^3 q + h_1\right).$$

On the other hand, we have $\dot{k_1} = 2h_1$ and $\dot{k_2} = 2h_2$ and $\dot{D} = \frac{k_1\dot{k}_1+\dot{k}_2}{D}$ so we can compute $\frac{d\mathcal{U}}{dt}$ as

$$\begin{aligned}\dot{\mathcal{U}} &= 6k_1^2\dot{k}_1 + 4\mu(k_1\dot{k}_2 + \dot{k}_1 + k_2) + 2(\mu - C)\dot{k}_1 + 4\mu\left(\frac{\dot{k}_1}{D} - \frac{k_1\dot{D}}{D^2}\right)\\ &= 12k_1^2h_1 + 8\mu(k_1h_2 + h_1k_2) + 4(\mu - C)h_1 + 8\mu(\frac{h_1}{D} + \frac{k_1^2h_1 + k_1h_2}{D^3}).\end{aligned}$$

Note that the first and third terms cancel immediately. Also, we compute $k_1h_2 + h_1k_2 = 2(pu^3 - qv^3)$ so the third terms also cancel. To check that the final terms cancel, we need to verify that

$$u^3p - 3uv^2p + 3u^2vq - v^3q + h_1 = h_1D^2 + k_1^2h_1 + k_1h_2.$$

3 Verifying connections with BVP

3.1 The Newton map without regularization

We first consider the case of an approximate connecting orbit which is homoclinic to an equilibrium. In this case, despite our mining procedure utilizing the regularization technique, the verification of a connecting orbit does not require any regularization. The appropriate BVP fomrulation is as follows.

Suppose $\hat{x}$ is a saddle-focus equilibrium for the CRTBP vector field, $f : \mathbb{R}^4 \to \mathbb{R}^4$ with a two-dimensional stable and unstable manifolds. We assume we have parameterized these manifolds. Specifically, we assume $P : \mathbb{C}^2 \to \mathbb{R}^4$ and $Q : \mathbb{C}^2 \to \mathbb{R}^4$ satisfying $P(\mathbb{D}^2) \subset W^u(\hat{x})$ and $P(\mathbb{D}^2) \subset W^s(\hat{x})$ and P, Q have been explicitly computed as high order Taylor expansions. Moreover, the Taylor coefficients for P, Q have the conjugate symmetry property as defined in **??** implying that a *real* local unstable manifold may be obtained explicitly as

$$W^u_{\text{loc}}(\hat{x}) = \left\{P\left(re^{i\theta}, re^{-i\theta}\right) : 0 \le r < 1, 0 \le \theta < 2\pi\right\}.$$

Moreover, we the boundary of the local unstable manifold is a 1-dimensional manifold with an explicit parameterization given by the formula,

$$P_\partial(\theta) = P(e^{i\theta}, e^{-i\theta}) \quad \theta \in [0, 2\pi).$$

Similarly, we have parameterizations for the local stable manifold and its boundary given by

$$\begin{aligned}W^s_{\text{loc}}(\hat{x}) &= \left\{Q\left(re^{i\theta}, re^{-i\theta}\right) : 0 \le r < 1, 0 \le \theta < 2\pi\right\}\\ Q_\partial(\theta) &= Q(e^{i\theta}, e^{-i\theta}) \quad \theta \in [0, 2\pi).\end{aligned}$$

Now, the Newton map for the multiple shooting scheme is defined to be $g : \mathbb{R}^{4n-1} \to \mathbb{R}^{4n}$ defined by

$$g(\alpha, x_1, \ldots, x_{n-1}, \beta, \tau) = (g_1, \ldots, g_n)$$

where, $\{\alpha, \beta, \tau\} \subset \mathbb{R}$, $\{x_j : 1 \le j \le n-1\} \subset \mathbb{R}^4$, and the coordinate functions are defined by

$$\begin{aligned}g_1(\alpha, x_1) &= \Phi\left(t_1, P_\partial(\alpha)\right) - x_1\\ g_j(x_{j-1}, x_j) &= \Phi\left(t_j, x_{j-1}\right) - x_j \qquad 2 \le j \le n-1\\ g_n(x_{n-1}, \beta, \tau) &= \Phi\left(\tau, x_{n-1}\right) - Q_\partial(\beta)\end{aligned}$$

for fixed constants $\{t_1, \ldots, t_{n-1}\}$.

We want to find a root of g via Newton's method. So, next we compute a formula for Dg. First, we let $M(j,k)$ denote the space of $j \times k$ matrices. We compute the derivative of each component function as block matrix elements in $M(4, 4n-1)$.

First block

$$Dg_1(\alpha, x_1) = \begin{pmatrix} \frac{\partial}{\partial \alpha}\Phi(t_1, P_\partial(\alpha)) & -\mathrm{I_d} & 0 \end{pmatrix}$$

where the first block is a column vector since

$$\frac{\partial}{\partial \alpha}\Phi(t_1, P_\partial(\alpha)) = D\Phi(t_1, P_\partial(\alpha)) \cdot DP(e^{i\alpha}, e^{-i\alpha}) \cdot \left(ie^{i\alpha}, -ie^{-i\alpha}\right) \in M(4,1)$$

and $I_d \in M(4,4)$ and the last block of zeros is in $M(4, 4n-6)$.

Middle blocks

For $1 < j < n$, we have

$$Dg_j(x_{j-1}, x_j) = \begin{pmatrix} 0 & D\Phi(t_j, x_{j-1}) & -\mathrm{I_d} & 0 \end{pmatrix}$$

where the first block of zeros is in $M(4, 4j-7)$ and the last block in $M(4, 4n-4j-2)$, and the two center blocks are in $M(4,4)$.

Last block

Finally, we have

$$Dg_n(x_{n-1}, \beta, \tau) = \begin{pmatrix} 0 & D\Phi(\tau, x_{n-1}) & f\left(\Phi(\tau, x_{n-1})\right) & -\frac{\partial}{\partial \beta}Q_\partial(\beta) \end{pmatrix}$$

where the first block of zeros is in $M(4, 4n-7)$, the second block in $M(4,4)$, the third block in $M(4,1)$, and the last block is a column vector since

$$\frac{\partial}{\partial \beta}Q_\partial(\beta) = -DQ(e^{i\beta}, e^{-i\beta}) \cdot (ie^{i\beta}, -ie^{-i\beta}) \in M(4,1)$$

3.2 The variational equation

In order to iterate Newton's method on g defined above, we need to compute expressions for the derivative of the flow, $D\Phi(t,x)$. This is obtained by solving the variational equation which requires computing the derivative of the vector field. We do this only for the original CR3BP since the numerical method will not require automatic differentiation. We write the CR3BP as a vector field, $f : \mathbb{R}^4 \to \mathbb{R}^4$ defined by

$$f(x,p,y,q) = \begin{pmatrix} p \\ 2q + (1-\mu)(x-\mu) + \mu(x+1-\mu) - \frac{(1-\mu)(x-\mu)}{d_1^3} - \frac{\mu(x+1-\mu)}{d_2^3} \\ q \\ -2p + y - \frac{(1-\mu)y}{d_1^3} - \frac{\mu y}{d_2^3} \end{pmatrix}$$

where

$$d_1 = \sqrt{(1-\mu)^2 + y^2} \qquad d_2 = \sqrt{(x+1-\mu)^2 + y^2}$$

We first compute partials for d_i which are given by

$$\begin{aligned}
\frac{\partial}{\partial x} d_1 &= \frac{x-\mu}{d_1} \\
\frac{\partial}{\partial y} d_1 &= \frac{y}{d_1} \\
\frac{\partial}{\partial x} d_2 &= \frac{x+1-\mu}{d_2} \\
\frac{\partial}{\partial y} d_2 &= \frac{y}{d_2}
\end{aligned}$$

Now, we have the partials for f_2, f_4 given by

$$\frac{\partial f_2}{\partial x} = 1 - (1-\mu)\left(\frac{1}{d_1^3} - \frac{3(x-\mu)^2}{d_1^5}\right) - \mu\left(\frac{1}{d_2^3} - \frac{3(x+1-\mu)^2}{d_2^5}\right)$$

$$\frac{\partial f_2}{\partial y} = \frac{\partial f_4}{\partial x} = 3y\left(\frac{(1-\mu)(x-\mu)}{d_1^5} + \frac{\mu(x+1-\mu)}{d_2^5}\right)$$

$$\frac{\partial f_4}{\partial y} = 1 - (1-\mu)\left(\frac{1}{d_1^3} - \frac{3y^2}{d_1^5}\right) - \mu\left(\frac{1}{d_2^3} - \frac{3y^2}{d_2^5}\right)$$

where the equality in the middle pair is due to the fact that the coordinates of f arise from partial derivatives of the potential, $\Omega \in C^2$. So in fact these coordinates are nothing more than the Hessian of Ω at $(x, y) \in \mathbb{R}^2$. Specifically, we write

$$D^2\Omega(x,y) := H = \begin{pmatrix} \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \\ \frac{\partial f_4}{\partial x} & \frac{\partial f_4}{\partial y} \end{pmatrix} = \begin{pmatrix} H_1 & H_2 \\ H_2 & H_3 \end{pmatrix}$$

With these computed, the Jacobian for f reads

$$Df(x,p,y,q) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ H_1 & 0 & H_2 & 2 \\ 0 & 0 & 0 & 1 \\ H_2 & -2 & H_3 & 0 \end{pmatrix}$$